Pedro Henrique Carvalho Alves

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Profile

I always search for increasing my knowledge about technology, especially in Quality Assurance area, software tests and automation tools for testing. I've implemented some automation processes in my previous companies like using pyautogui to automate Desktop Applications, which made the Desktop automation faster and more user friendly for even non-technical users, Jenkins integrated with Selenium + Python to run tests in a regular basis and send emails including attached reports, Postman
+ Newman integration with Python Tkinter APPs to run API tests and generate HTML reports through the APP.

Work Experience

CoForge – April 2026 – Present –
Associate Engineering – QA

- Working with Xray to run the test executions, create test plans and test cases.
- Designed and implemented automated tests on pytest to validate an OS project which includes a Kubernetes environment with Clusters, Pods, Pvcs and other Kubernetes Kinds.
- Contributed to Open Dev automated tests
- Implemented an AI knowledge comprised of many agents that allow to create test plans, test cases and test executions on Jira as well as trigger Jenkins jobs to run the automation scripts.
- Worked with a Kubernetes based OS, which is composed by different apps and ran test cases on Kubernetes. For instance, by creating PVCs pointing out to different storage classes, Pods with data inside mounting on the PVCs/Pvs and ensuring that system was stable by stressing the Kubernetes components as well as validating across different K8s versions.
- Continuos use of AI, especially Kiro from Amazon to speed up and enhance the daily activities by writing automation scripts, using the AI knowledge base to help with questions about the system and using a learn agent to store all the important information internally for the team.
- Measuring the credit efficiency of different AI internal systems by using Pandas to create graphs and tables to choose the best tools when it comes to credit spending.
- Integrated different MCP tools with our IDE to help with the Jira efficiency on looking up tickets, creating new test cases as well as inserting new information on Confluence. On top of that, some ssh mcp tools were also integrated to help with the connection to the labs and on executing K8s commands.

Integritas Inc – March 2023 – April 2026
Full SDET

- Designed and implemented CI/CD pipelines in Jenkins to automatically execute regression suites, reducing manual regression execution time by approximately 40–60%.
- Automating tests using Selenium with Python and pyTest to write the test cases and generate the reports after the executions.
- Architected an internal automation tool using **Python + Tkinter**, enabling non-technical stakeholders to execute API and UI tests through a user-friendly interface, increasing adoption of

automated testing across teams.

- Execute Postman collections using newman in order to validate the API.
- Developed SQL validation queries to ensure correct database transactions for POST/GET endpoints, improving backend data consistency.
- WEB Bank APP tests using Postman for the backend side and Selenium with Python for the front end.
- Developed SQL validation queries to ensure correct database transactions for POST/GET endpoints, improving backend data consistency.
- Browser Stack to test the mobile app using mobile emulators.
- Developed SQL validation queries to ensure correct database transactions for POST/GET endpoints, improving backend data consistency.
- Writing Documentation for the Software in Confluence.
- Tests in AWS using AppSync and Postman to validate the APIs
- DynamoDB tables to make sure we have the registers saved there
- CloudWatch to monitor the logs and trace the execution.
- Playwright with Javascript to automate tasks by writing Playwright scripts to upload thousands of documents as well as to inject some Javascript code through the page.evaluate method available in Playwright syntax

Tech Skills used: *Tech Stack: Python, Selenium, PyTest, Playwright (JS), Postman, Newman, Jenkins, AWS (AppSync, DynamoDB, CloudWatch), SQL, JIRA, XRAY, Confluence, BrowserStack.*

INATEL – March 2021 - March 2023
Full SDET

- Cypress Automation for a WEB platform responsible for adding totems to control IoT devices through areas. Cypress automation was used to automate the Login process, Registration and also to ByPass some screens by using the cache and local storages.
- Desktop APP testing using pyautogui to automate the scenarios. Manual testing using JIRA and Zephyr Scale. Implementing threads to monitor different processes through execution.
- Supporting QA teams using Cypress and helping them with Cypress's Javascript automation. Building pipelines with UI tests in Cypress as well as some API tests and mock endpoints in Cypress to have the API responses and payloads.
- Support other team members by teaching them how to execute tests using Postman + Newman and how to obtain the HTML reports from the tests.
- Measurement of the performance using JMeter.
- Bug reports using Postman and validating the endpoints through postman.
- Confluence pages to document the software processes, scenarios, and test credentials.
- Jenkins to execute the API tests.
- Newman + Python to execute the API tests on the desktop platform.
- Version control using BitBucket and git commands.

Tech Skills used: *Postman, Python, Cypress, pyautogui, Newman, Jenkins, BitBucket, JMeter, XRAY, ZephyrScale, JIRA, Python Threads, Pytest*

Riversoft – November 2019 - March 2021
Junior SDET

- Start-up company which did not have a QA department at that time, so they allowed me to start new manual QA processes as well as automation using Robot Framework.
- WEB Platform, which consists of a place to transfer files safely using layers of cryptography.
- Implementation of the QA department from scratch by using Zephyr Scale to write the test cases
- Confluence documentation with the API endpoints to validate over the platform.
- Test cases for Ubuntu and Windows to validate different OS.
- Test reports using newman and Python to trace down the bugs and test executions.
- Jenkins Pipelines to enhance the process and automatically execute the API validation.
- Version control using GitHub.

Tech Skills used: *RobotFramework, Python, Newman, Jenkins, GitHub, ZephyrScale, JIRA*

Achievements

- API built using FASTAPI + Python to save information about soccer leagues, teams and players. API documentation can be found at https://soccer-api-jzbt.onrender.com/docs
- Personal Project to build an Amazon mock using Javascript and Asynchronous programming (async await). In this project I learned how to deal with big JSON files, local

storage in the browser, and the main Javascript concepts and functions like filter, map and reduce. https://github.com/PedroCarvalho1500/AMAZON_JAVASCRIPT

- Task management project built with Javascript + Postgres. Throughout this project, I could increase my SQL skills (using the Postgres database) as well as improve my Javascript skills and how to present the information in a HTML page. https://github.com/PedroCarvalho1500/TASKS_MANAGEMENT_EJS
- Full UI automation using Robot Framework. I attained that during my first job in which they allowed me to learn how to automate using Robot Framework from scratch. I automated the UI features and common user actions. This reduced the number of manual tests we required and also could find some bugs in advance once we had the new mergers. **Technologies used: Robot Framework, Python**
- I built a Python Tkinter APP in which we could execute our API tests and UI tests and see the reports generated in our browser. It allowed us to execute our tests in a User-friendly way instead of going to the terminal and typing commands. This also helped to present the automation to the end users. The UI shows some options and once the user chooses one, it will automatically execute the tests. **Technologies used: Python, Tkinter, sqlite3**
- Python terminal menus to execute our API testing and execution of API tests in parallel. This helped reduce the time taken to finish the execution. Instead of executing each foolder of the API, it allows you to execute more than one API folder at the same time. **Python, Linux Shell Script.**
- Designed a Jenkins pipeline to automatically execute the API Postman collections using newman and the UI tests in Selenium with Python. **Technologies used: Python, Jenkins, Selenium with Python, Postman, Newman**

Education

National Institute of Telecommunications – INATEL – Computer Engineering, Graduated in 2019, July.

IFSULDEMINAS – Câmpus Machado, IT Technic, graduated in 2013.

Languages

Portuguese – Native Language

English – Advanced 1 (C1) certificate by English Live. CEFR level C1.
https://www.efset.org/cert/sjVzGx

Education and Courses

- Test Automation with Robot Framework - Basic, spent time - 04 hours. https://udemy-certificate.s3.amazonaws.com/image/UC-062d4edd-6baf-43b5-83ee-e4976d7d04f2.jpg?v=1603884953000

- Automation of API Tests with Postman + Test Project - January 3, 2021, spent time of 09 hours. http://ude.my/UC-907ffc54-5f4e-43df-8781-cd09e89875f8

- Python PCEP-30-02 certification from Python Institute, May/2025 - Certification can

be found at https://verify.openedg.org/?id=FNtk.sY8a.ai0X

- Learn Selenium with Python, PyTest + Live Project – 46.5 hours. https://www.udemy.com/certificate/UC-ef00b3ea-45fe-4269-a2df-978336b0dfc2/

- Scientific Article published in INCITEL congress, held in 2018, July, from 4 to 6. https://inatel.br/incitel/documents/2018.pdf Article Title: Desenvolvimento de interface grafica para jogo de reabilitacão de membro superior

- 100 Days of Code: The Complete Python Pro Bootcamp for 2023 – 58 horas. https://www.udemy.com/certificate/UC-78e51082-453e-4c4b-ac35-c1fa9ba7c4a8/. Projects developed through this course: https://github.com/PedroCarvalho1500/bootcamp_python

Volunteer Work

- IT Volunteer Teacher in Casa Viva Inatel , a middle school maintained by INATEL university.